Janssen effect

Part A. Hydrostatics (0.5 points)

A.1 (0.2pt)

$p_1 =$

A.2 (0.3pt)

$p(z) =$

Part B. Janssen effect in a cylindrical vessel with vertical walls (2.9 points)

B.1 (0.5pt)

Equation:

B.2 (1.2pt)

$A =$

$B =$

$\lambda =$

B.3 (1.2pt)

$\rho =$

$K =$

Part C. Conical vessel (2.3 points)

C.1 (0.1pt)

$r(z) =$

C.2 (0.6pt)

Equation:

C.3 (0.8pt)

$A =$

$\gamma =$

$\beta =$

C.4 (0.8pt)

Graph:

$\frac{P_{\text{max}}}{\rho g z_{\text{max}}} =$

Part D. What if the vessel is lifted? (1.5 points)

D.1 (0.4pt)

Equation:

D.2 (0.3pt)

$A =$

$B =$

$\lambda =$

D.3 (0.8pt)

$H\ =$

Part E. Fluid outflow through the wall (1.8 points)

E.1 (0.2pt)

Equation:

E.2 (0.1pt)

$W_1 =$

E.3 (0.3pt)

$A\ =$

E.4 (0.3pt)

$p_2 =$

E.5 (0.3pt)

$v(h)\ =$

E.6 (0.6pt)

$\tau =$

Part F. "Flow" of granules through a wall (1 point)

F.1 (0.2pt)

$\theta_r =$

F.2 (0.8pt)

$q\ =$

$n\ =$

Air shower

Part A. Electromagnetic shower (2.0 points).

A.0 (0.2pt)

$d_{\mathrm{e}} =$

A.1 (0.4pt)

$n_{\max} =$

A.2 (0.4pt)

$l_{\mathrm{e}} =$

A.3 (0.2pt)

$N_{\max} =$

A.4 (0.8pt)

$Q_{\mathrm{e}} =$

Part B. Hadronic shower (2.0 points).

B.1 (0.4pt)

$n_{\mathrm{hadr}}^{\max} =$

B.2 (1.0pt)

$n_{\mathrm{EM}}^{\max} =$

B.3 (0.6pt)

$\beta_{\mathrm{miss}} =$

Part C. Air shower (3.0 points).

C.1 (1.3pt)

$h_{\mathrm{stop}} =$

C.2 (0.5pt)

$\beta^{\mathrm{EM}} =$

C.3 (0.2pt)

$R_{\mathrm{stop}} =$

C.4 (0.2pt)

$S =$

C.5 (0.8pt)

$\Delta t =$

Part D. Data Analysis (3.0 points).

D.1 (1.0pt)

$\kappa =$

D.2 (0.5pt)

$E_0 =$

D.3 (1.5pt)

$\rho_{\mathrm{atm}} =$

PN junctions in semiconductors

Part A. Electrostatics (5.0 points)

A.1 (0.5pt)
Relation:

A.2 (0.5pt)
Proof:

A.3 (1.0pt)

$E_x(x) =$

A.4 (1.0pt)

$\varphi(x) =$

A.5 (0.5pt)
Plot:

A.6 (1.0pt)

(Formula) $d_n =$ (Value) $d_n =$

(Formula) $d_p =$ (Value) $d_p =$

A.7 (0.5pt)

(Formula) $E_{\max} =$ (Value) $E_{\max} =$

Part B. Junction with external voltage (5.0 points)

B.1 (0.5pt)

$n_1 =$

$p_1 =$

B.2 (1.0pt)
Equation:

B.3 (0.6pt)

(Formula) $L_p =$ (Value) $L_p =$

B.4 (0.8pt)

$p_1(x) =$

B.5 (0.8pt)

$j_p =$

B.6 (0.8pt)

$I =$

B.7 (0.3pt)
Plot:

B.8 (0.2pt)

$I_\infty =$

$I_1 =$